

بڑے پروگرام — مکمل کوڈ پریکٹس: Topic 15

1 پروگرام: Student Grade Management System

```
def get_grade(marks):
    if marks >= 80: return "A"
    elif marks >= 70: return "B"
    elif marks >= 60: return "C"
    elif marks >= 50: return "D"
    else: return "F"

students = []
n = int(input("کتنے طلباء: "))
for i in range(n):
    name = input(f"نام {i+1} طالب علم: ")
    marks = int(input(f"{name} کے نمبر: "))
    students.append({"name": name, "marks": marks, "grade": get_grade(marks)})

print("\n--- نتائج ---")
print(f"{'15>:نام':<15} {'10>:نمبر':<10} {'گریڈ':<10}")
print("-" * 35)
for s in students:
    print(f"{s['name']:<15} {s['marks']:<10} {s['grade']}<10}")

with open("results.txt", "w") as f:
    for s in students:
        f.write(f"{s['name']}: {s['marks']} - Grade {s['grade']}\n")
print("\nresults.txt میں محفوظ ہو گئے نتائج")
```

2 پروگرام: Simple ATM Banking System

```
class BankAccount:
    def __init__(self, owner, balance=0):
        self.owner = owner
        self.balance = balance
        self.history = []

    def deposit(self, amount):
        if amount <= 0:
            print("رقم درست نہیں")
```

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        return

    self.balance += amount
    self.history.append(f"جمع: {amount}")
    print(f"{amount} روپے جمع ہوئے۔ بقیہ {self.balance}")

def withdraw(self, amount):
    if amount <= 0:
        print("رقم درست نہیں")
    elif amount > self.balance:
        print("ناکافی رقم")
    else:
        self.balance -= amount
        self.history.append(f"نکاسی: {amount}")
        print(f"{amount} روپے نکلے۔ بقیہ {self.balance}")

def show_statement(self):
    print(f"\n--- {self.owner} کا بیان ---")
    for record in self.history:
        print(record)
    print(f"موجودہ بقیہ: {self.balance}")

account = BankAccount("Ahmad", 1000)
while True:
    print("\n1. جمع 2. نکاسی 3. بیان 4. خروج")
    choice = input("انتخاب: ")
    if choice == "1":
        account.deposit(int(input("رقم: ")))
    elif choice == "2":
        account.withdraw(int(input("رقم: ")))
    elif choice == "3":
        account.show_statement()
    elif choice == "4":
        print("خدا حافظ!")
        break

```

3 پروگرام: Word Frequency Analyzer

```

import string

def clean_word(word):
    return word.lower().strip(string.punctuation)

def analyze_text(text):
    words = text.split()

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frequency = {}
for word in words:
    word = clean_word(word)
    if word:
        frequency[word] = frequency.get(word, 0) + 1
return frequency

def top_words(frequency, n=5):
    sorted_words = sorted(frequency.items(), key=lambda x: x[1], reverse=True)
    return sorted_words[:n]

print("دو بار دبائیں (Enter متن لکھیں):")
lines = []
while True:
    line = input()
    if line == "":
        break
    lines.append(line)

text = " ".join(lines)
freq = analyze_text(text)

print(f"\nکل منفرد الفاظ: {len(freq)}")
print(f"\nسب سے زیادہ استعمال شدہ الفاظ:")
for word, count in top_words(freq):
    print(f" {word}: {count} بار")

```